BROADER ECONOMY IN TRANSITION · BASE SCENARIOS (SSP/RCP)

What are the environmental consequences of micro- and meso-scale decisions?
in consumption, process design and operation, in diverse contexts and over the transition

which units operate?

how they operate?

Pillar I · Market consequences
Which suppliers respond, when and where, with what environmental profile

Pillar II · Process consequences
How processes and their environmental flows respond to design, operation and context

Common strategy
local markets and processes as instances of broader ones; simplifications tested

PhD I — Task 1.1
Building a framework for scenario-consistent market responses, from model perturbation to heuristics

PhD II — Task 2.1
Consolidating process knowledge in an interoperable framework: models of any sophistication, set by scenario and decision

PhD I — Task 1.1
Testing marginal-supplier heuristics against structural model responses

Energy
Pillar I case

PhD III — Task 2.2
AI-assisted model building, evaluated where the answer is known

MSc I — Task 1.2
Testing heuristics against past metal capacity, then deriving marginal supply from dynamic MFA

Metallurgy
shared case

MSc II — Task 2.3
Calibrating engineering models against reported plant performance, and testing what generic datasets miss

MSc III — Task 1.3
Disaggregating scenario regions to the local markets where decisions play out

Fertilizers
shared case

PhD III — Task 2.2
Representing an emerging process before deployment, as alternative configurations

TYPE OF REFERENCE

Errors vs perturbable models (counterfactual)

Errors vs observed and reported data (retrospective)

Sensitivity where no reference exists

OUTCOMES

Scenario-consistent market responses
and the errors of common heuristics, characterized

Cross-case synthesis
which simplifications matter, and where

Interoperable process-model classes
and an evaluated workflow to build more

TIMING
Years of activity, 1 to 5
1 2 3 4 5

Consistent, discriminating prospective consequential LCA for environmental decisions across contexts
with the decision cost of common simplifications characterized